



Jeremías Likerman

ADJUNCT RESEARCHER, CONICET · ASSOCIATE PROFESSOR, UNIVERSITY OF BUENOS AIRES
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Education

PhD, 2015	Geological Sciences, University of Buenos Aires (UBA)	Argentina
Thesis: <i>Prediction of fracturing on irregular geological surfaces using static methods</i> . Supervisor: Prof. Ernesto Cristallini.		
BSc, 2010	Geological Sciences, University of Buenos Aires (UBA)	Argentina

Main Career History

May 2026–present	Associate Professor, Department of Geological Sciences, UBA	Argentina
Teaching: Geostatistics and Geoinformatics.		
2018–present	Adjunct Researcher (Investigador Adjunto), CONICET-IDEAN	Argentina
Structural geology, geomechanics, geodynamics, natural fractures, and geothermal systems. Research career from 2018; promoted to Investigador Adjunto, November 2023.		
May–Sep 2025	Visiting Researcher, GEO3BCN–CSIC	Spain
Aug–Sep 2024	Visiting Researcher, GEO3BCN–CSIC	Spain
2015–2017	Postdoctoral Researcher, CONICET and Universitat Politècnica de Catalunya	Argentina/Spain
2013–2026	Teaching appointments, Department of Geological Sciences, UBA	Argentina
2013–2014	Visiting Doctoral Researcher, Stanford University	USA

Major Research Achievements

- More than 15 years of research in structural geology and multiscale numerical modelling, integrating field observations, analogue experiments, geomechanics, and 2D/3D simulations.
- Developed predictive approaches for natural-fracture distribution and connectivity in folded structures and energy reservoirs, including the Yacoraite carbonates and the Vaca Muerta Formation.
- Leads and co-leads national and international projects on subduction dynamics, lithospheric deformation, natural fractures, and geothermal systems; coordinator of CONICET–CSIC cooperation on geothermal modelling (2024–2026).
- Supervises early-career researchers and undergraduate, doctoral, and postdoctoral work in structural geology, geomechanics, and numerical modelling.

Selected Recent Major Publications

1. Villarroel, M.A., et al., including **Likerman, J.** (2026). The role of compressional structures in the development of collapse calderas: Insights from analogue models. *Journal of Volcanology and Geothermal Research*, 473, 108582.
2. Gianni, G.M., Gallo, L.C., **Likerman, J.**, et al. (2025). Slab underthrusting is the primary control on flat-slab size. *Science Advances*, 11(28).
3. Correa-Luna, C., Yagupsky, D.L., **Likerman, J.**, & Barcelona, H. (2025). Impact of large-scale structures on fracture network connectivity: Insights into the Vaca Muerta unconventional play. *Tectonophysics*, 230640.
4. Plotek, B., **Likerman, J.**, & Cristallini, E. (2024). Geomechanical modeling of fault-propagation folds: A comparative analysis of finite-element and the trishear kinematic model. *Journal of Structural Geology*, 105064.
5. Navarrete, C., et al., including **Likerman, J.** (2024). Massive Jurassic slab break-off revealed by a multidisciplinary reappraisal of the Chon Aike silicic large igneous province. *Earth-Science Reviews*, 249, 104651.
6. Gianni, G.M., **Likerman, J.**, Navarrete, C.R., Gianni, C.R., & Zlotnik, S. (2023). Ghost-arc geochemical anomaly at a spreading ridge caused by supersized flat subduction. *Nature Communications*, 14, 2083.

Full Curriculum Vitae

Professional Profile

Research profile: Geoscientist specializing in structural geology, geomechanics, and multiscale numerical modelling, with more than 15 years of experience in research, university teaching, fieldwork, and hands-on student training. His research addresses geodynamic, geothermal, and geomechanical processes in Andean systems and energy reservoirs, integrating two- and three-dimensional models to study subduction zones, crustal deformation, volcanic systems with geothermal potential, and natural and induced fractures, with emphasis on the Vaca Muerta Formation.

Teaching, outreach, and knowledge transfer: Teaching experience in Geological Surveying, Geostatistics, and Introduction to Geology, including practical work in geological mapping, spatial analysis, geological interpretation, geospatial tools, UAVs, and quantitative methods. Participation in public science communication, STEAM education, and community initiatives addressing territorial and urban-environmental issues and access to public services.

Academic and Postdoctoral Training

Postdoctoral Fellowship

CONICET AND UPC

Argentina/Spain

2015–2017

- Research topic: thermal evolution of oceanic lithosphere. Supervisors: Dr. Ernesto Cristallini (CONICET-UBA) and Dr. Sergio Zlotnik (UPC).

Visiting Researcher

STANFORD UNIVERSITY

USA

2013

- Courses in Structural Geology and MATLAB applied to Geosciences.

PhD in Geological Sciences

UNIVERSITY OF BUENOS AIRES

Argentina

2010–2015

- Thesis: *Prediction of fracturing on irregular geological surfaces using static methods*. Supervisor: Dr. Ernesto Cristallini.

BSc in Geological Sciences

UNIVERSITY OF BUENOS AIRES

Argentina

2005–2010

- Thesis: *Geology and structure of the Tres Cruces compound anticline, Jujuy, Argentina*. Supervisor: Dr. Ernesto Cristallini.

Appointments and Professional Experience

CURRENT APPOINTMENTS

Adjunct Researcher (Investigador Adjunto)

CONICET-IDEAN

Argentina

2018–present

- Research areas: fracture prediction, geodynamic modelling, and geothermal-potential modelling. CONICET research career from 2018; promoted to Investigador Adjunto, November 2023.

Associate Professor

DEPARTMENT OF GEOLOGICAL SCIENCES, UNIVERSITY OF BUENOS AIRES

Argentina

May 2026–present

- Courses: Geostatistics and Geoinformatics.

Institutional Coordinator

IDEAN, CONICET-UBA

Argentina

Sep 2025–present

- Council Member since April 2024.

PREVIOUS TEACHING APPOINTMENT

Teaching Assistant

DEPARTMENT OF GEOLOGICAL SCIENCES, UNIVERSITY OF BUENOS AIRES

Argentina

2013–2026

- Courses: Introduction to Geology, Geological Surveying, and Geostatistics.

PROFESSIONAL COURSES AND FIELD ACTIVITIES

Geology of Unconventional Reservoirs

INSTRUCTOR IN CHARGE

Argentina

2025

- Professional training course focused on the geological characterization of shale reservoirs.

Advanced Research Field Trip

FIELD INSTRUCTOR

Argentina

2025

- Organizer and instructor for *Perspectives on Vaca Muerta: subsurface attributes analysed from surface exposures*.

RESEARCH VISITS

Visiting Researcher

GEO3BCN-CSIC

Spain
May-Sep 2025

- Thermomechanical modelling of volcanic zones for geothermal-potential assessment. Host: Dr Ivone Jiménez-Munt.

Visiting Researcher

GEO3BCN-CSIC

Spain
Aug-Sep 2024

- Development of coupled numerical models for volcanic and geothermal systems.

Visiting Researcher

BARCELONATECH (UPC)

Spain
2015, 2017, 2019, 2022

- Research stays at the Numerical Computation Laboratory.

Research Projects

2025	Geothermal potential of volcanic zones (Principal Investigator). AECT-2025-1-0009	Spain
2025	Structural controls in volcanic systems: magmatic intrusions and damage-zone development associated with hydrocarbon generation (Co-PI). Williams Foundation	Argentina
2024	Assessment of geothermal potential in volcanic zones using thermomechanical numerical models (PI). LINCGL2024	Spain
2024	Geodynamic modelling of subduction zones and mantle plumes (PI). AECT-2024-2-0027	Spain
2023	Geodynamics of the Adria microplate, from mantle to surface (Co-PI). GEO3BCN	Spain
2022	Geodynamic modelling of subduction zones (PI). AECT-2022-3-0023	Spain
2020	Landscape dynamics and geological hazards associated with dams in mountain regions (Co-PI). PICT-2019-04011	Argentina
2020	Analogue and numerical modelling: contributions to tectonic and structural problems (Co-PI). PICT-2019-00997	Argentina
2018	Natural fracturing and first-order structures: predictive models applied to Cerro Domuyo (Co-PI). UBACyT	Argentina
2017	Geological evolution of the Andes and its economic and environmental impact (Co-PI). PUE 22920160100051	Argentina
2016	Numerical modelling of the thermal evolution of oceanic lithosphere (PI). PICT 2015-1226	Argentina
2016	Application of analogue and numerical models to tectonic-structural problems (Co-PI). PICT-2016-1407	Argentina

Research Supervision

RESEARCHER SUPERVISION

Plotek, Berenice

ASSISTANT RESEARCHER

Argentina
Since Aug. 2025

- Supervisor: Likerman, J.; co-supervisor: Cristallini, E.

POSTDOCTORAL SUPERVISION

Plotek, Berenice

FAULT-PROPAGATION FOLDS: A MECHANICAL APPROACH USING FINITE-ELEMENT NUMERICAL MODELLING

Argentina
2024

- Supervisors: Cristallini, E. and Likerman, J.

DOCTORAL SUPERVISION

Correa Luna, Clara

NATURAL FRACTURING IN THE VACA MUERTA FORMATION: CHARACTERIZATION AND MODELLING

Argentina
2023

- Supervisors: Yagupsky, D. and Likerman, J.

Undergraduate Thesis Supervision

Chandía Sepúlveda, Alexander Ignacio

TECTONIC INVERSION OF A METAMORPHIC COMPLEX IN THE DESEADO MASSIF: INSIGHTS FROM ANALOGUE MODELLING

Argentina
In progress

- Supervisors: Likerman, J. and Navarrete, C.

De Rosa, Tomás

QUATERNARY DEFORMATION IN THE FIAMBALÁ VALLEY, CATAMARCA

Argentina
In progress

- Supervisors: Likerman, J. and Sagripanti, L.

Blanco, Lucía

GEOHERMAL-ELECTRIC POTENTIAL OF THE CERRO BLANCO VOLCANIC COMPLEX: INFLUENCE OF RHEOLOGY AND STRUCTURE THROUGH NUMERICAL MODELS

Argentina
In progress

- Supervisors: Likerman, J. and Lamberti, C.

Celis Santisteban, Celso Angel

STRUCTURAL EVOLUTION OF AN INVERTED METAMORPHIC COMPLEX IN THE DESEADO MASSIF: INSIGHTS FROM NUMERICAL MODELLING

Argentina
In progress

- Supervisors: Likerman, J. and Navarrete, C.

Ruiz, Luciana

ASSESSMENT OF THE GEOHERMAL-ELECTRIC POTENTIAL OF THE CERRO BLANCO VOLCANIC COMPLEX USING NUMERICAL MODELLING

Argentina
2026

- Supervisors: Lamberti, C. and Likerman, J.

Gramajo, Pilar

QUATERNARY DEFORMATION IN THE CHASCHUIL VALLEY, WESTERN SIERRA DE LAS PLANCHADAS, CATAMARCA

Argentina
2025

- Supervisors: Likerman, J. and Sagripanti, L.

Acosta, Luana

GEOMORPHOLOGY OF THE RÍO GRANDE VALLEY BETWEEN BARDAS BLANCAS AND PORTEZUELO DEL VIENTO, MENDOZA

Argentina
2023

- Supervisors: Winocur, D. and Likerman, J.

Mazzini, Martín

GEOLOGY OF THE EASTERN SLOPE OF CERRO PERITO MORENO, EL BOLSÓN, RÍO NEGRO

Argentina
2022

- Supervisors: Likerman, J. and Tobal, J.

Relañez, Gastón

STRUCTURE AND NEOTECTONICS OF THE TABLADITAS AREA, JUJUY, ARGENTINA

Argentina
2019

- Supervisors: Yagupsky, D. and Likerman, J.

Ventisky, Esteban

GEOLOGY, STRUCTURE, AND FRACTURING OF THE LAJAS FORMATION IN THE COVUNCO ANTICLINE, NEUQUÉN

Argentina
2017

- Supervisors: Likerman, J. and Tobal, J.

Correa Luna, Clara

STRUCTURAL GEOLOGY AND FRACTURING IN THE TRES CRUCES BASIN, JUJUY

Argentina
2015

- Supervisors: Yagupsky, D. and Likerman, J.

Peer-Reviewed Publications

The role of compressional structures in the development of collapse calderas: Insights from analogue models

Villarroel, M. A., Browning, J., Jara, P. P., Harnett, C. E., Holohan, E. P., Marquardt, C. J., Guzman, M., **Likerman, J.**
Journal of Volcanology and Geothermal Research (2026) p. 108582. Elsevier, 2026

Impact of large-scale structures on fracture network connectivity: Insights into the Vaca Muerta unconventional play, Neuquén basin, Argentina

Correa-Luna, C., Yagupsky, D. L., **Likerman, J.**, Barcelona, H.
Tectonophysics (2025) p. 230640. Elsevier, 2025

Slab underthrusting is the primary control on flat slab size

Gianni, G. M., Gallo, L. C., **Likerman, J.**, Echaurren, A., Gianni, C. R., Faccena, C.
Science Advances 11.28 (2025). American Association for the Advancement of Science, 2025

Massive Jurassic slab break-off revealed by a multidisciplinary reappraisal of the Chon Aike silicic large igneous province

Navarrete, C., Gianni, G., Tassara, S., Zaffarana, C., **Likerman, J.**, Márquez, M., Wostbrock, J., Planavsky, N., Tardani, D., Frassetto, M. P.
Earth-Science Reviews 249 (2024) p. 104651. Elsevier, 2024

Geomechanical modeling of fault-propagation folds: A comparative analysis of finite-element and the trishear kinematic model

Plotek, B., **Likerman, J.**, Cristallini, E.
Journal of Structural Geology (2024) p. 105064. Elsevier, 2024

Ghost-arc geochemical anomaly at a spreading ridge caused by supersized flat subduction

Gianni, G. M., **Likerman, J.**, Navarrete, C. R., Gianni, C. R., Zlotnik, S.
Nature communications 14.1 (2023) p. 2083. Nature Publishing Group UK London, 2023

Natural fracture characterization of the Los Catutos Member (Vaca Muerta Formation) in the southern sector of the Sierra de la Vaca Muerta, Neuquén Basin, Argentina

Correa-Luna, C., Yagupsky, D. L., **Likerman, J.**, Barcelona, H.
Journal of South American Earth Sciences 120 (2022) p. 104081. Elsevier, 2022

Kinematics of fault-propagation folding: Analysis of velocity fields in numerical modeling simulations

Plotek, B., Heckenbach, E., Brune, S., Cristallini, E., **Likerman, J.**
Journal of Structural Geology 162 (2022) p. 104703. Elsevier, 2022

The influence of climate on the dynamics of mountain building within the Northern Patagonian Andes

García Morabito, E., Beltrán-Triviño, A., Terrizzano, C. M., Bechis, F., **Likerman, J.**, Von Quadt, A., Ramos, V. A.
Tectonics 40.2 (2021) e2020TC006374. Wiley Online Library, 2021

The effects of small-scale convection in the shallow lithosphere of the North Atlantic

Likerman, J., Zlotnik, S., Li, C.-F.
Geophysical Journal International 227.3 (2021) pp. 1512–1522. Oxford University Press, 2021

Fracture network analysis of Yacoraite Formation in the Tres Cruces sub-basin, northwestern Argentina

Luna, C. C., Yagupsky, D. L., **Likerman, J.**
Journal of South American Earth Sciences (2019) p. 102446. Elsevier, 2019

Cenozoic intraplate tectonics in Central Patagonia: Record of main Andean phases in a weak upper plate

Gianni, G. M., Echaurren, A., Folguera, A., **Likerman, J.**, Encinas, A., García, H. P. A., Dal Molin, C., Valencia, V. A.
Tectonophysics 721 (2017) pp. 151–166. Elsevier, 2017

Closure type effects on the structural pattern of an inverted extensional basin of variable width: Results from analogue models

Jara, P., **Likerman, J.**, Charrier, R., Herrera, S., Pinto, L., Villarroel, M., Winocur, D.
Journal of South American Earth Sciences (2017). Elsevier, 2017

Climatic and Tectonic forcing on alluvial fans in the Southern Central Andes

Terrizzano, C. M., Morabito, E. G., Christl, M., **Likerman, J.**, Tobal, J., Yamin, M., Zech, R.
Quaternary science reviews 172 (2017) pp. 131–141. Elsevier, 2017

Structural and statistical characterization of joints and multi-scale faults in an alternating sandstone and shale turbidite sequence at the Santa Susana Field Laboratory: Implications for their effects on groundwater flow and contaminant transport

Cilona, A., Aydin, A., **Likerman, J.**, Parker, B., Cherry, J.
Journal of Structural Geology 85 (2016) pp. 95–114. Elsevier, 2016

Middle to late Miocene extensional collapse of the North Patagonian Andes (41° 30'–42° S)

Tobal, J. E., Folguera, A., **Likerman, J.**, Naipauer, M., Sellés, D., Boedo, F. L., Ramos, V. A., Gimenez, M.
Tectonophysics 657 (2015) pp. 155–171. Elsevier, 2015

Geodynamic context for the deposition of coarse-grained deep-water axial channel systems in the Patagonian Andes

Ghiglione, M. C., **Likerman, J.**, Barberón, V., Giambiagi, L. B., Aguirre-Urreta, B., Suarez, F.
Basin Research 26.6 (2014) pp. 726–745. Wiley Online Library, 2014

Role of basin width variation in tectonic inversion: insight from analogue modelling and implications for the tectonic inversion of the Abanico Basin, 32°–34° S, Central Andes

Jara, P., **Likerman, J.**, Winocur, D., Ghiglione, M. C., Cristallini, E. O., Pinto, L., Charrier, R.

Along-strike structural variations in the Southern Patagonian Andes: Insights from physical modeling

Likerman, J., Burlando, J. F., Cristallini, E. O., Ghiglione, M. C.

Tectonophysics 590 (2013) pp. 106–120. *Elsevier*, 2013

Fellowships and Distinctions

2027	Research Fellowship in Japan Kobe University	<i>JSPS</i>
2025	SZNet 2025 Chile Field Trip National Science Foundation	<i>SZ4D</i>
2022	Academic Mobility Fellowship among Partner Institutions Ibero-American University Association for Postgraduate Studies	<i>AUIP</i>
2016	International Research Stay National Scientific and Technical Research Council	<i>CONICET</i>
2015	Postdoctoral Fellowship National Scientific and Technical Research Council	<i>CONICET</i>
2013	Doctoral Research Stay at Stanford University Fulbright – Bunge & Born Foundation	<i>Doctoral fellowship</i>
2013	Doctoral Fellowship Type II National Scientific and Technical Research Council	<i>CONICET</i>
2010	Doctoral Fellowship Type I National Scientific and Technical Research Council	<i>CONICET</i>

Skills

Programming: $\text{\LaTeX}$, Python, R, MATLAB, HTML5.

Modelling tools: Underworld2, LaMEM, Golem, ASPECT; analogue modelling, experimental design, material properties, PIV, and post-processing; HPC workflows.

Spatial analysis: QGIS, GMT, PostGIS; seismic interpretation, geological mapping, and UAV-assisted field methods.

Fieldwork: more than 15 years of experience, extensive knowledge of the Andes, campaign planning and logistics, and team leadership.

Languages: Spanish (native), English (fluent, C2), Italian (A2), Catalan (A2).